Graphic Era HILL UNIVERSITY
Established by an Act of the State Legislature of Uttarakhand (Adhiniyam Sankhya 12 of 2011)
University under section 2(f) of UGC Act, 1956

End Term (Odd) Semester Examination November 2025

Roll no.....

Name of the Course and semester: B.Tech CSE V
Name of the Paper: Computer Networks-1
Paper Code: TCS-514
Time: 3 hour

Maximum Marks: 100

Note:

(i) All the questions are compulsory.
(ii) Answer any two sub questions from a, b and c in each main question.
(iii) Total marks for each question is 20 (twenty).
(iv) Each sub-question carries 10 marks.

Q1. (2X10=20 Marks)

a. Discuss Circuit switching and Packet switching with the help of diagrams. Give at least 5 differences among these. Also, explain types of packet switching in brief. CO1

b. A packet of size 2,000 bytes is sent over three links with transmission rates of 5 Mbps, 10 Mbps, and 20 Mbps. Compute:
i. Transmission delay on each link
ii. Total transmission delay
iii. If propagation delay per link is 5 ms, calculate the end-to-end delay. CO1

c. Explain the layered architecture of the Internet protocol stack. Describe the functions of each layer and compare it with the OSI reference model. Also discuss why the TCP/IP model became more widely adopted than OSI. CO1

Q2. (2X10=20 Marks)

a. Describe DNS Query resolving. State Iterative and Recursive DNS resolving mechanisms with the help of proper diagram and steps. Also state DNS caching concept in brief. CO2

b. A non-persistent HTTP connection is used to download a web page that includes 1 base HTML file and 6 images. Assume:
i. Each object requires 1 RTT for request–response
ii. TCP connection setup requires 1 RTT
iii. Transmission time is negligible
Calculate the total time required to download the entire page using:
i. Non-persistent HTTP without parallel connections
ii. Persistent HTTP (without and with pipelining) CO2

c. What is P2P file distribution? In a BitTorrent P2P file distribution, a file of size 400 MB is shared among N = 20 peers.
Given:
i. Each peer has an upload rate of 2 Mbps
ii. The original server has an upload rate of 5 Mbps
iii. Peers upload as soon as they receive pieces
iv. BitTorrent distributes file pieces in parallel
Using the P2P file distribution, Compute the minimum time required to distribute the file to all peers. CO2

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Q3. (2X10=20 Marks)

a. Differentiate between connection-oriented and connectionless services in the Transport Layer. Discuss how these two approaches affect reliability, ordering, congestion control, and overhead. CO3

b. Station A needs to send a message consisting of 9 data packets to station B using a sliding window (window size =3) and Go-Back-N error control strategy. All packets are ready and immediately available for transmission. If every 5th data packet that A transmits gets lost (but no ACKs from B ever get lost), then what is the number of packets that A will transmit for sending message to B. CO4

c. Describe in detail the UDP segment structure. Explain the purpose of each field and discuss why UDP includes a checksum even though it is a connectionless protocol. Also state how checksum is used to handle errors. CO3

Q4. (2X10=20 Marks)

a. Explain the process of TCP Connection Establishment and termination using the three-way handshake and four-step connection teardown. Discuss sequence number and acknowledgement number values with the help of an example. CO3

b. Consider a Selective Repeat sliding window protocol that uses a frame size of 1KB to send data on a 1.5Mbps link with one-way latency of 50msec. To achieve a link utilization of 60%, find the minimum number of bits required to represent the sequence number field. CO4

c. State the TCP Congestion Control Mechanisms in detail:
   i. Slow Start
   ii. Congestion Avoidance
   iii. Congestion Detect

   Describe how each stage modifies the congestion window and ensures network stability. CO4

Q5. (2X10=20 Marks)

a. Explain IPv4 Header with the help of suitable diagram. State the significance of fields present in the IPv4 Header structure. Also, compare these to IPv6 Header in brief. CO5

b. Given the IPv4 address 172.16.35.200/20, answer the following:
   i. Find the network address of the subnet containing this host.
   ii. Find the broadcast address for this subnet.
   iii. Calculate the number of usable hosts in this /20 subnet.
   iv. Determine the range of usable host addresses for this subnet.

   Present the answers in dotted-decimal notation and show how you determine the asked values. CO5

c. Answer the followings: CO5
   i. Discuss the limitations of Classful addressing and describe how CIDR overcomes these limitations.
   ii. State the major steps involved in transitioning from IPv4 to IPv6